

Cake Cutting

Properties of the Cut and Choose Protocol

Today, we'll focus primarily on fairness properties.

Suppose two people need to split a heterogeneous and divisible good. The usual euphemism in the fair division literature is that of cutting a cake. More practically, the good could be an estate (e.g., in a divorce settlement) or processing time on a computer cluster (perhaps with some times of the day more valuable than others).

Why not just split the good 50/50? This makes sense when the good is homogeneous, but it's not clear what this means with a heterogeneous good. A player may value a part of the good much more than another, and different players can have different opinions about which parts are the most valuable.

We all know a reasonable protocol for two-person cake-cutting.

The Cut and Choose Protocol

1. Player 1 splits the good into two pieces A and B , such that the player's value for each is exactly half that of the entire good.
2. Player 2 picks whichever of A, B she likes better.

The description above is the intended behavior of the players in the protocol—we'll talk shortly about whether or not they are incentivized to follow this behavior. We've been led all our lives to believe that this is a "fair" protocol. But is it? How would we formally argue one way or the other?

Here's the formal model. The good, or "cake," is the unit interval $[0, 1]$. (Yes, it's a weird-looking cake.) Each player i has a valuation function v_i , which specifies the value $v_i(S)$ to i of a given subset S of the cake. Values are nonnegative, and we assume that $v_i([0, x])$ is continuous in x , so a cut giving player i half her value exists. Pieces may be unions of intervals, and we allocate the entire cake.

1. We normalize each v_i such that $v_i([0, 1]) = 1$, their value for the whole cake. This is more or less without loss of generality, by scaling.
2. v_i is additive on disjoint subsets. That is, if $A, B \subseteq [0, 1]$ are disjoint, then

$$v_i(A) + v_i(B) = v_i(A \cup B).$$

Exercise. Player 1 values the cake uniformly. Player 2 values only the left quarter, uniformly within that quarter. Run cut-and-choose with player 1 cutting at $1/2$. What does each player receive, and how much does she value it?

Player 2 chooses the left half and receives value 1. Player 1 receives the right half, worth $1/2$ to her. Equal physical sizes need not mean equal values.

Is the cut and choose protocol strategyproof?

Here, ask whether either player can gain by departing from the prescribed behavior while the other follows it.

For the second player: since they can't affect the split of the cake into A and B , and they are supposed to choose the piece they like better, they have no incentive to deviate.

For the first player: if they know something about the second player's valuation function, they might want to deviate from the protocol. For example, suppose the good is a hot fudge sundae, the first player likes all parts of the sundae equally, while the second player likes ice cream but really cares about the cherry. The first player could split the sundae into the cherry and the rest, knowing that the second player would take the cherry, leaving a very valuable piece for the first player.

If the first player doesn't know anything about what the second player wants, and assumes that the second player will always leave the piece that is worse for the first player, then the first player is incentivized to follow the protocol (to guarantee herself a piece with value $\frac{1}{2}$). Thus the prescribed behavior is not a dominant strategy for the cutter.

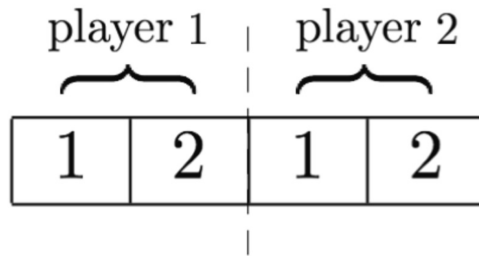
In the exercise above, player 1 can instead cut at $1/4$. Player 2 still chooses the left piece, worth 1 to her; player 1 now gets value $3/4$ instead of $1/2$. Player 2's valuation and choice rule have stayed fixed. The cutter's action changed.

An allocation is *Pareto optimal* if there is no other allocation that makes both players weakly better off and at least one strictly better off.

Is the cut and choose protocol guaranteed to produce a Pareto optimal solution (assuming both players behave as intended)?

Consider four equal quarters. Player 1 values only quarters 1 and 3, at $1/2$ each; player 2 values only quarters 2 and 4, at $1/2$ each. Compare cutting into two halves with another allocation.

Cutting into two halves gives each player value $1/2$, whichever half player 2 chooses. Giving quarters 1 and 3 to player 1 and quarters 2 and 4 to player 2 gives both value 1. Both strictly improve, so the cut-and-choose outcome is not Pareto optimal. The alternative uses disconnected pieces, which our model allows.



Can we come up with a protocol that is both Pareto optimal and strategyproof for both players?

For this question, suppose both players value every set of positive length strictly positively.

Give the entire cake to player 1, independently of anything the players say or do. Neither player can change her allocation, so this is strategyproof. Giving player 2 positive value requires transferring a set of positive length, which strictly hurts player 1. Thus the outcome is Pareto optimal under the stated assumption. It is not what we mean by “fair.” Without that assumption, transferring a piece that only player 2 values could be a Pareto improvement.

What about “fairness?” Equally happy?

Does cut-and-choose give both players the same value for their own pieces?

No. The first player gets a piece that she values at $1/2$, while the second player may receive value greater than $1/2$. In the first exercise the values were $1/2$ and 1 .

One definition of “fairness” is that each player receives at least her fair share, at least from her perspective.

Definition 1. An allocation $A_1, \dots, A_n$ of the entire cake to n players is *proportional* if $v_i(A_i) \geq 1/n$ for each player i .

A second definition is that no player wants to trade places with the other player.

Definition 2. An allocation $A_1, \dots, A_n$ is *envy-free* if $v_i(A_i) \geq v_i(A_j)$ for each pair i, j .

This compares both pieces using the same player’s valuation.

Does cut-and-choose satisfy these properties? How are they related for general n , and for two players?

Assume both players follow the protocol.

The cutter values each piece at $1/2$. The chooser takes her favorite piece, whose value is at least $1/2$ because her values for the two pieces sum to 1 . Thus cut-and-choose is proportional. Neither player envies: the cutter is indifferent and the chooser gets her favorite.

For n players, envy-freeness implies proportionality: for a fixed player i , all n bundles are worth at most $v_i(A_i)$ to her, and their values sum to 1. Hence $1 \leq nv_i(A_i)$.

For any allocation of the entire cake to two players, additivity gives $v_i(A_j) = 1 - v_i(A_i)$. Therefore

$$v_i(A_i) \geq v_i(A_j) \iff v_i(A_i) \geq 1 - v_i(A_i) \iff v_i(A_i) \geq 1/2.$$

Thus proportionality and envy-freeness are equivalent for two players. The dictator allocation fails both: player 2 gets value 0 and values player 1's piece at 1.

Fair Division of Indivisible Goods

Exercise. Two players both value a single indivisible good at 1. Allocate the good. Can the allocation be envy-free or proportional?

Whoever does not get the good receives value 0 and values the other player's bundle at 1. Thus no allocation of the good is envy-free or proportional. Leaving goods unallocated is not allowed.

Definition 3. We have a set $N = [n]$ of players and a set $\mathcal{M}$ of m goods. Each player i has a nonnegative, additive valuation:

$$v_i(S) = \sum_{g \in S} v_i(g).$$

An allocation $A = (A_1, \dots, A_n)$ is a partition of $\mathcal{M}$ into disjoint bundles, one per player. Empty bundles are allowed; all goods are allocated.

For the numerical examples, we leave the values unnormalized. Proportionality then asks for $v_i(A_i) \geq v_i(\mathcal{M})/n$.

Notions of Envy Freeness

Our definition of envy-freeness is the same as before. Since it is not always possible to achieve it, we relax the condition.

Definition 4. For an allocation $A = (A_1, \dots, A_n)$:

1. The allocation is *envy-free up to one good* (EF1) if for every i, j with $A_j \neq \emptyset$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for some } g \in A_j.$$

2. The allocation is *envy-free up to any good* (EFX) if for every i, j ,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \quad \text{for every } g \in A_j.$$

An empty bundle is never envied. Removing a good is only a comparison: the actual allocation still contains all the goods. Here EFX uses every good, including zero-valued goods.

How do the quantifiers differ? Which condition is stronger?

EF1 allows a different witness good for each ordered pair (i, j) ; EFX requires the inequality for every good in the envied bundle. Thus

$$\text{EF} \implies \text{EFX} \implies \text{EF1}.$$

The first implication uses nonnegative values. For the second, choose any good from a nonempty bundle; an empty bundle causes no envy.

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References

Tim Roughgarden. CS269I: Incentives in Computer Science, 2016.